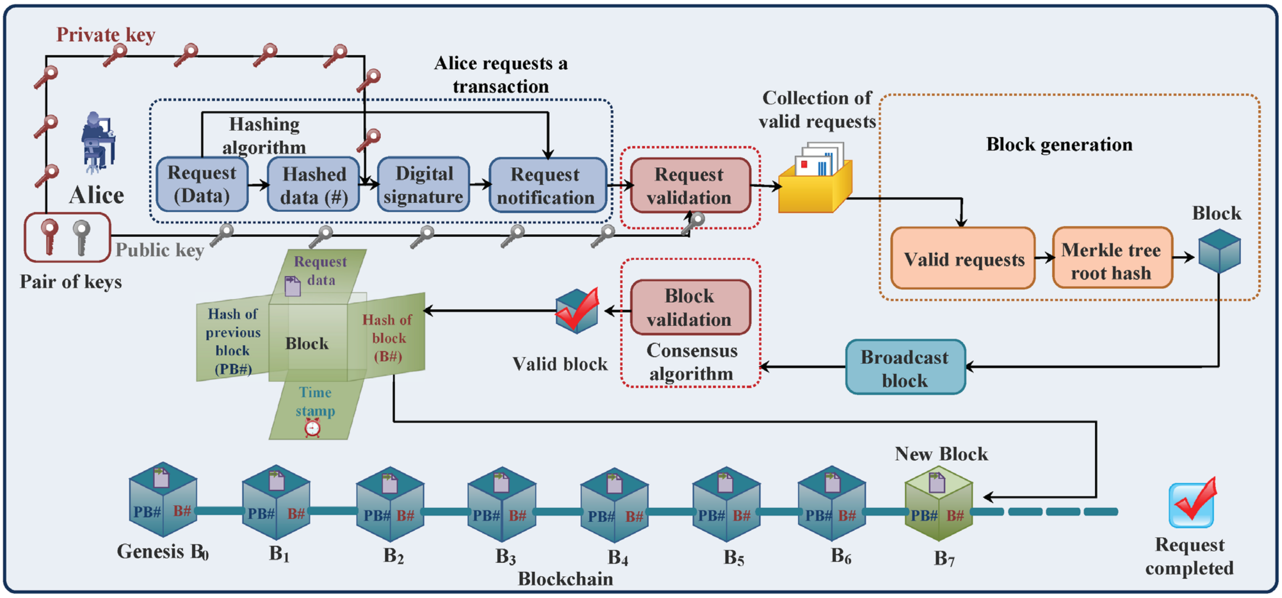

Figure 3: Quantum-secured blockchain

brute force methods that need 2^n operations to compromise an n-bit hash function, Grover's algorithm lowers the complexity to around 2^(n/2) operations. What this implies is that cryptographic hash algorithms such as SHA-256, which are deployed in blockchain systems currently, can be susceptible to collision attacks. This means the transaction can be meddled with or cryptographic hashes can be reverse-engineered.

Key recovery attacks and digital wallet vulnerability: Another critical vulnerability is key recovery attacks. Quantum computers may extract private keys from public keys, rendering entire blockchain wallets completely compromised. Since the security of most crypto wallets depends on protection by means of asymmetric cryptographical techniques, this vulnerability can potentially expose hundreds of billions worth of digital content to unauthorised entities. Quantum technology-enabled malicious users can, in theory, steal money, impersonate valid users, and alter transaction histories.

Post-quantum cryptography (PQC) for blockchain security

Quantum computing is no longer something on the horizon — blockchain networks need to get ready for it now. If action isn't taken, decentralised security may be jeopardised, ruining the future of Web3, DeFi, and blockchain apps.

While the use of hybrid cryptographic methods, which blend both classical and quantum-resistant algorithms, could be a transitional security measure, blockchain networks will have to adopt post-quantum cryptography (PQC) if they are to remain secure and valid. PQC algorithms are used to provide quantum attack-resistance, thereby ensuring the security of transactions, smart contracts, and dApps based on blockchain. Several PQC methods are already available for this purpose.

Lattice-based cryptography: CRYSTALS-Kyber and CRYSTALS-Dilithium are instances of lattice-based cryptographic primitives that rely on the computational intractability of lattice problems, which can withstand attacks from even Shor's quantum computers. These schemes can potentially be used in blockchain key exchange protocols, digital signatures, and identity verification systems.

```
from pycryptodome.publickey import NTRU
key = NTRU.generate(1024)
message = b"Quantum-safe blockchain"
ciphertext = key.encrypt(message)
decrypted_message = key.
decrypt(ciphertext)
print("Original:", message)
print("Decrypted:", decrypted_message)
```

Hash-based cryptography: Blockchain networks also depend on cryptographic hash functions in terms of consensus and immutability. Hash-based signature schemes such as XMSS (eXtended Merkle Signature Scheme) and SPHINCS+ are quantum-resistant and enable the usability of digital signatures deployed in blockchain transactions. Hash-based cryptography is particularly appropriate for transaction authentication as well as ledger protection in blockchain networks. Here is an example of a post-quantum hash-based signature (SPHINCS+) implemented in Python: